✓ EXPERIMENT NO 6

VOLTAGE REGULATION OF ALTERNATOR

AIM

To predetermine the voltage regulation of the given 3 phase alternator by

i) emf method ii) mmf method and iii) Zero power factor (Potier) method.

APPARATUS

S.No.	Name of the apparatus	Type	Range	Ouantitv
1.	Voltmeter			
2.	Voltmeter			
3.	Voltmeter			
4.	Ammeter			
5.	Ammeter			
6.	Ammeter			
7.	Ammeter			
8.	Rheostat			
9.	Rheostat			
10.	Rheostat			
11.	Tachometer			

PRINCIPLE

The terminal voltage of an alternator under load conditions is different from the open circuit voltage due to the effects of armature resistance, lakage reactanee and armature reaction. Voltage regulation is defined as the rise in voltage, expressed as per cent of rated voltage, when the load current is reduced to zero, the field excitation and frequency being maintained constant. Thus,

Voltage regulation$=\frac{E_f-V}{V}\times 100$

The term rise in voltage used in the above definition pre-supposes a resistive or inductive load. If the load is capacitive, the magnetizing effect of armature reaction, due to the leading current, may cause V to be higher than E_f, thus causing a drop in voltage, when the load current is reduced to zero. In that case, the regulation is negative.

The regulation of a synchronous generator can be predetermined by the following methods: a) synchronous impedance or emf method, b) mmf or ampere turn method c) zero power factor or potier method.